

CHE 454. MOLECULAR & MACROMOLECULAR ENGINEERING

Instructor: Naveen Kumar Dandu

Term: Spring 2023

Course Description:

Advanced course in polymer science and engineering. Polymerization, polydispersity, molecular configuration, solution properties, thermodynamics, glass and rubbery states, crystallization, viscoelasticity, elastic properties, multiphase systems. Course Information: 3 undergraduate hours. 4 graduate hours.

Prerequisite(s):

Knowledge of concepts studied in fundamental science & engineering courses, and mathematical skills equivalent to chemical engineering/mechanical engineering/physics undergraduate education is presumed. Familiarity with calculus, ordinary differential equations and vector analysis will be assumed.

Textbooks & Reading Materials:

No specific textbook is required. Lecture notes, assignments and extra reading material discussed/detailed in the class is sufficient. List of books followed for the course are:

- Polymer Physics by Michael Rubinstein and Ralph Colby, Oxford, 2003.
- R. J. Young and P. A. Lovell. Introduction to Polymers. 3rd edition, CRC Press, 2011.
- Soft Matter Physics by Masao Doi, Oxford Univ. Press, Oxford, 2013.
- The Physics of polymers by Gert Strobl, Springer, 3rd edition, 2007.
- Intermolecular and Surface Forces by Jacob Israelachvili, Academic Press, 3rd edition, 2011.

Grading:

Assignments and exams grades are done fairly and as quickly as possible. Some partial credit will be given to incomplete or less-than-perfect solutions; a generic break-up of partial credit will be outlined in solutions provided for assignments and mid-terms.

10 Homework Assignments (20% of the grade)

1 Mid-Term Exam in mid-March (40%), TBD

1 Term paper (40%) - Writeup 30 and Presentation 15 (more details are discussed in class)

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Topics covered in class

- + Soft matter physics
- + Introduction to Polymers/Macromolecules
- + Ideal and Real chains
- + Molecular and Macromolecular Thermodynamics: solutions and blends
- + Networks and Gelation
- + Rheology
- + Classical Statistical Mechanics
- + Surfactants, Lipids and Liquid Crystals
- + Presentations and Term papers as Finals